# Derivatives and Roots

## Purpose

The purpose of this lab is for you to acquire an understanding of the relationships among a function, its first and second derivatives, and the roots of these functions. You will apply this understanding to critical analysis of non-mathematical statements.

## Pre-Lab

Go to GeoGebra Graphing Calculator. This lab will largely take place using this app. Every member of the group should have the page open and be working to follow the lab on their own laptop. Each group will be responsible for all members being able to use the app and get the lab done.

1. Graph the function
$$f(x) = x^4 - 4,056x^2 + 16,192x + 2,234,247$$
on GeoGebra (just type it into an input box[1]). Set the viewing window to horizontal range $[-70, 70]$ and vertical range $[-3,000,000, 2,500,000]$ by clicking the Settings (gear) icon in the top right, and entering the values under *Dimensions* (`xMin`, `xMax`, `yMin`, `yMax`)[2].

2. GeoGebra can find the roots and extrema of the function for you! In the Algebra panel on the left, click the three vertical dots (⋮) next to your formula (or click the formula) and select *Special Points*. This will add the roots, extrema, and $y$-intercept to the graph and display their coordinates in the Algebra panel. You can delete the $y$-intercept (click its three dots and select *Delete*), as we will not . be using that.

3. Our next step is to superimpose the graph of $f'$ on the graph of $f$. Compute the derivative of $f(x)$ by hand, then add it to your graph. When you're done, your screen should show the graphs of both $f$ and $f'$.

4. Just as you did with $f$, add in the roots and extrema of $f'$ by selecting *Special Points* from its three-dots menu (⋮), and delete the $y$-intercept.

[1] If typing on your keyboard doesn't do anything, click the keyboard icon at the bottom left and use it to input.

[2] Note: Enter numbers without commas in GeoGebra, e.g., `-3000000` and `2500000`. Alternatively, you can type `ZoomIn(-70, -3000000, 70, 2500000)` into the input bar.

## Part I: The First Derivative

**Questions in Parts I and II of the lab should be done on paper and written in a Google Doc. No Geogebra until Part III!**

5. Describe the relationship between the roots of $f'$ and the graph of $f$? Use the concept of tangent lines to explain your observation.

6. Describe the relationship between the sign of $f'$ and the graph of $f$. Explain your observation in the context of rates of change and values of $f$.

7. What is the relationship between the locations of the roots of $f$ and the locations of the roots of $f'$?

8. Consider the function $g(x) = f(x) - 2,500,000$. How many roots does this function have? What is the derivative of $g(x)$?

9. Using $g(x)$ instead of $f(x)$, answer the same questions you did in question 7.

10. Use your conclusions from questions 7 and 9 to make general statements about the relationship between the number of roots of a function $h$ and the number of roots of $h'$:

    - If a differentiable function $h$ has roots located at $r_1$ and $r_2$, what can you conclude about the behavior of $h'$ between $r_1$ and $r_2$?
    - If $h'$ has exactly $k$ roots, how many roots could $h$ have?
    - If $h$ has exactly $k$ roots, how many roots could $h'$ have?

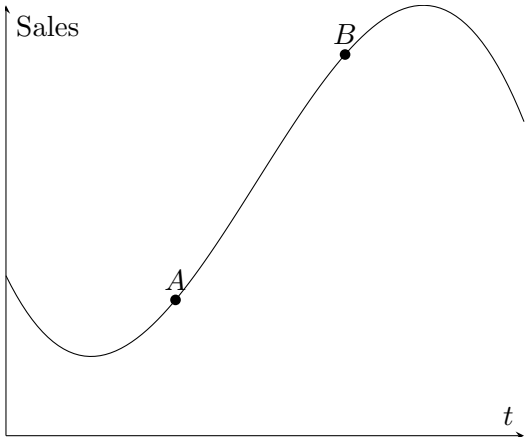
11. Complete the statement of each of the following theorems.

    **Maximum/Minimum Value Theorem:** If the function $f$ has a local minimum or a local maximum at $x = c$, and if $f'(c)$ exists, then $f'(c) =$ ____

    **Rolle's Theorem:** If $f(r_1) = 0$ and $f(r_2) = 0$ and if $f$ is differentiable on the interval $[r_1, r_2]$, then $f'$ has _____ on that interval.

## Part II: Concavity and Inflection Points

Suppose the curve below gives the daily sales totals for a new item in your department store. Consider the points $A$ and $B$ on the curve:




12. If you look closely at a small region on the curve around each of the points $A$ and $B$, you will see that although the rates of increase of sales at these two points are the same, there is a difference in the shape of the curve at the two points. Can you describe the difference?

13. Hopefully, you used the phrases "concave up" and "concave down" in answering the previous question. State a definition or a characterization of the phrases "concave up" and "concave down" sufficient to enable someone who knows no calculus to use them correctly in describing features of a graph.

14. You will notice that this graph changes concavity between points $A$ and $B$. As you may know, the point at which this change occurs is called an inflection point, which we will designate by "IP." On the diagram above, indicate the location of the IP and note that the curve "flexes" at that point.

15. Where on the sales curve above is the tangent line the steepest? Describe the trend of the slopes of tangent lines just before and just after the point that you've chosen.

## Part III: Derivatives and Inflection Points

16. We will now turn our attention back to the function $f$ which you graphed, along with its derivative function, in the Pre-Lab. On the curve representing $f$ locate the IPs as best you can just by looking (no need to write down any coordinates. Just look to see if you can see approximately where the IPs are).

17. Describe the relationship between the location of the IPs of $f$ and the graph of $f'$? Explain why this relationship should be true.

18. Your graph already includes the locations of the extrema of $f'$. Explain how to use this information to find the IPs of $f$. Using this, calculate the coordinates of the IPs of $f$.

19. Calculate $f''$ by hand and add it to your graph. Add its roots to the graph as well (using *Special Points*, then delete its extrema and $y$-intercept).

20. Switch to the Tools panel on the left by clicking the Tools icon  (click *MORE* at the bottom of the panel if needed to expand all tools). Under the *Lines* section, select the Line tool  , and draw vertical lines connecting the extrema of $f'$ to the roots of $f''$. How could you use these lines to find the IPs of $f$?

21. In the Tools panel, select the Intersect tool  (under *Basic Tools* or *Points*), then click each vertical line and the graph of $f$ to locate the IPs of $f$ exactly (hint: your answer to the previous question should help). Compare your answers to the ones from question 18.

## Report

Your lab report should consist of answers to the questions below, written neatly on a separate piece of paper. Each lab group will submit one report.

We did not cover the concept of *differentiability* this semester. It hasn't been a big deal so far. Turns out though, I kind of lied to you in this lab.

You see, it's not always true that between any two roots of a function there's at least one root of the derivative. There's a really important exception, which this report explores.

1. Consider this piecewise linear function:

$$f(x) = \begin{cases} x & 0 \leq x \leq 2 \\ 2 - x & 2 < x \leq 4 \end{cases}$$

   (a) Draw a graph of this function.

   (b) Fill in the blanks $f(x)$ has ___ roots. Between them, it has a _____.

   (c) Find $f'(x)$: first for $0 \leq x < 2$, then for $2 < x \leq 4$, then fill in the blanks:

$$f'(x) = \begin{cases} ____ & 0 \leq x < 2 \\ _____ & 2 < x \leq 4 \end{cases}$$

   (d) Draw the graph of the function.

   (e) What is $f'(0)$? (Hint: this is a bit of a trick....)

> The function $f(x)$ in this question is not *differentiable* at $x = 2$. This is the exception to the rule that between any two roots of a function there is at least one root of the derivative. The function goes from increasing to decreasing at $x = 2$, but its derivative is never zero. Instead, it sharply jumps from positive to negative.

2. Draw a graph of a discontinuous function $f(x)$ that has two roots, but whose derivative $f'(x)$ has no roots. Explain why this is possible.

> Two ways in which functions can fail to be differentiable are discontinuity and sharp turns. In this question, the function is discontinuous, so it is not differentiable. Therefore, it is possible for the function to have two roots, but for its derivative to have no roots.

3. If possible, draw graphs of differentiable functions whose domains are $(-\infty, \infty)$ and that have the following characteristics. If it is not possible, explain why. Make sure that it is clear what the behavior of the function is as $x \to \infty$ and $x \to -\infty$.

(a) $f'$ has four distinct zeros and $f$ has one zero.

(b) $f'$ has one zero and $f$ has four distinct zeros.

(c) $f(x) < 10$ for all $x$, $f'$ has one zero and $f$ has one zero.

(d) $f'$ has two distinct zeros and $f$ has no zeros.

4. If possible, draw graphs of continuous functions whose domains are $(-\infty, \infty)$ and that have the following characteristics. If it is not possible, explain why. Make sure that it is clear what the behavior of the function is as $x \to \infty$ and $x \to -\infty$.

   (a) $f(a) = 0$ and $f(b) = 0$ but $f'(x) \neq 0$ for any $x$ between $a$ and $b$.

   (b) $f(a) = 0$ and $f(b) = 0$ (these are the only two zeros of $f$) and $f'(x) = 0$ for exactly two values between $a$ and $b$.

5. Give formulas for functions whose domains are $(-\infty, \infty)$ with the following characteristics:

   (a) $f$ is continuous at 2, has a local minimum at 2, but $f'(2) \neq 0$.

   (b) $f$ is continuous at 0, $f'(0) = 0$, but $f$ has neither a maximum nor a minimum at 0.

   (c) $f$ is continuous at 0, $f'(0) = 0$, $f''(0) \neq 0$, but $f$ does not have an inflection point at 0.

   (d) $f$ is differentiable, $f'$ has two distinct zeros and $f$ has two distinct zeros.

**Rolle's Theorem**

If $f(r_1) = 0$ and $f(r_2) = 0$ and if $f$ is differentiable on the interval $[r_1, r_2]$,
then $f'$ has _____ on that interval.

6. Explain in detail why the differentiability of a function is important in the context of the relationship between the roots of a function and the roots of its derivative. Use examples to illustrate your explanation.